



Differences in Visual Search and Choice Depending on Item Familiarity

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ABSTRACT

Familiarity affects visual search and choice behavior.

- ▶ When choosing between two items, subjects look back and forth between the options while making their decision.
- ▶ Time spent inspecting each option correlates with what the subject chooses.
- ▶ The visual search and decision is further impacted by the familiarity of the option set.

MOTIVATIONS

- ▶ In life we choose between both familiar and novel options.
- ▶ It is difficult to control for subjective familiarity of everyday objects in decision-making studies that examine the cognitive bases of such decisions.
- ▶ In this experiment we manipulate familiarity while subjects learn the value of abstract stimuli.
- ▶ We compare choice and visual search patterns when subjects are choosing between more or less familiar stimuli.

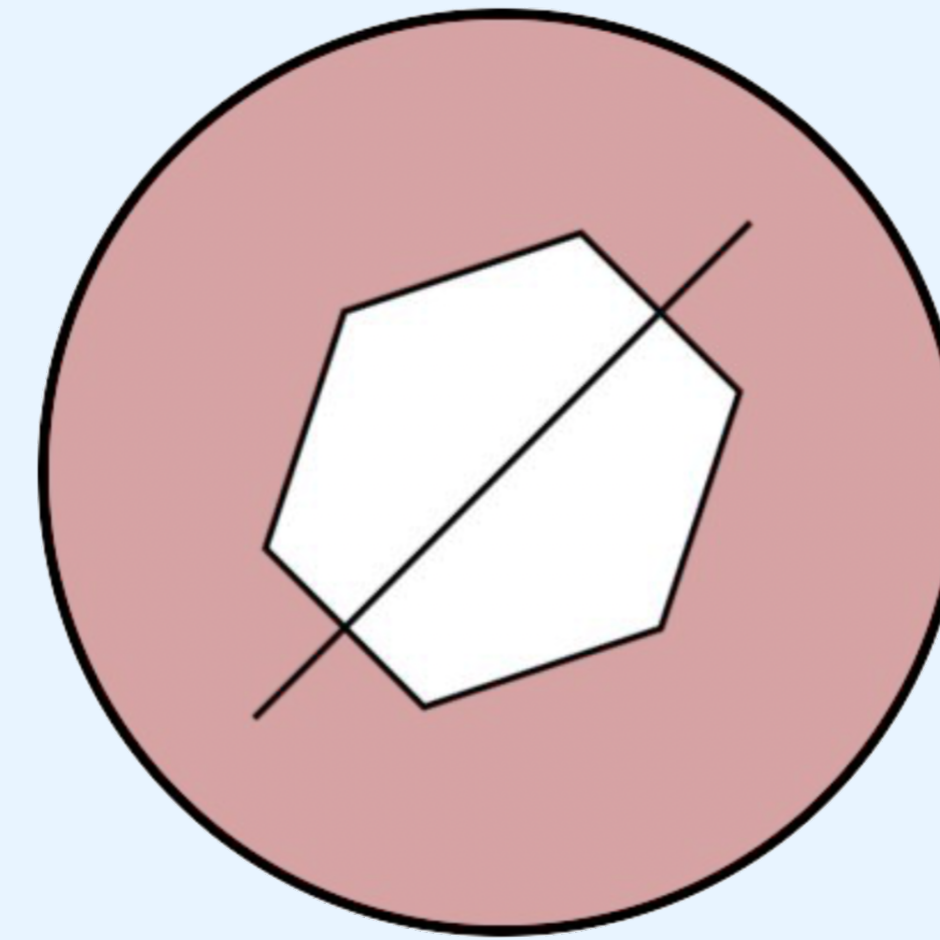
BACKGROUND

- ▶ Neuroeconomics describes how value-based decisions are made and relates cognitive processes underlying these decisions to neurobiological correlates.
- ▶ Previous research established that attention has a direct effect on what people choose [1] [2] [3].
- ▶ However, most of this work examines choice between real-world items (e.g. snack foods) without controlling for subjects' familiarity with the options.
- ▶ It is currently not known whether the cognitive processes and the neural correlates of value-based choice depends on familiarity with item values.
- ▶ Here we examine if choice and attention differ when we manipulate how well subjects know the value of the items they are choosing between.

FAMILIAR VERSUS NOVEL ITEMS IN REAL WORLD

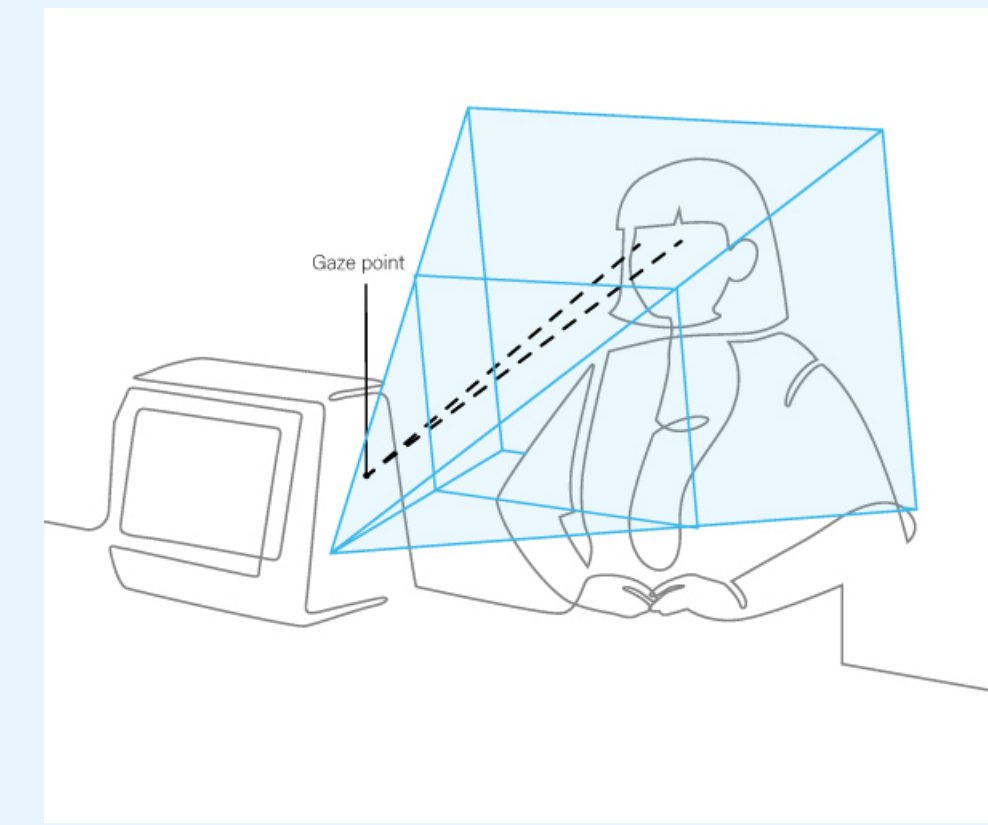
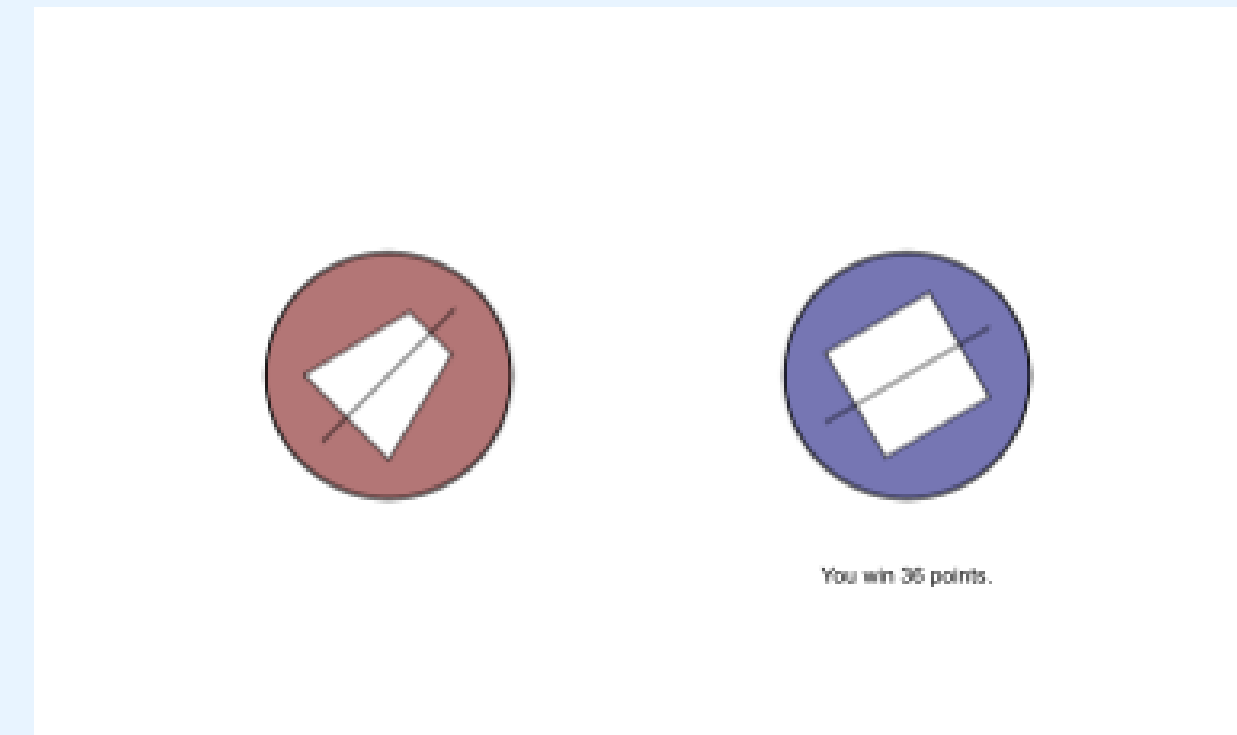


METHODS



Stimulus attributes:

1. Central Shape
2. Background Color
3. Line Orientation



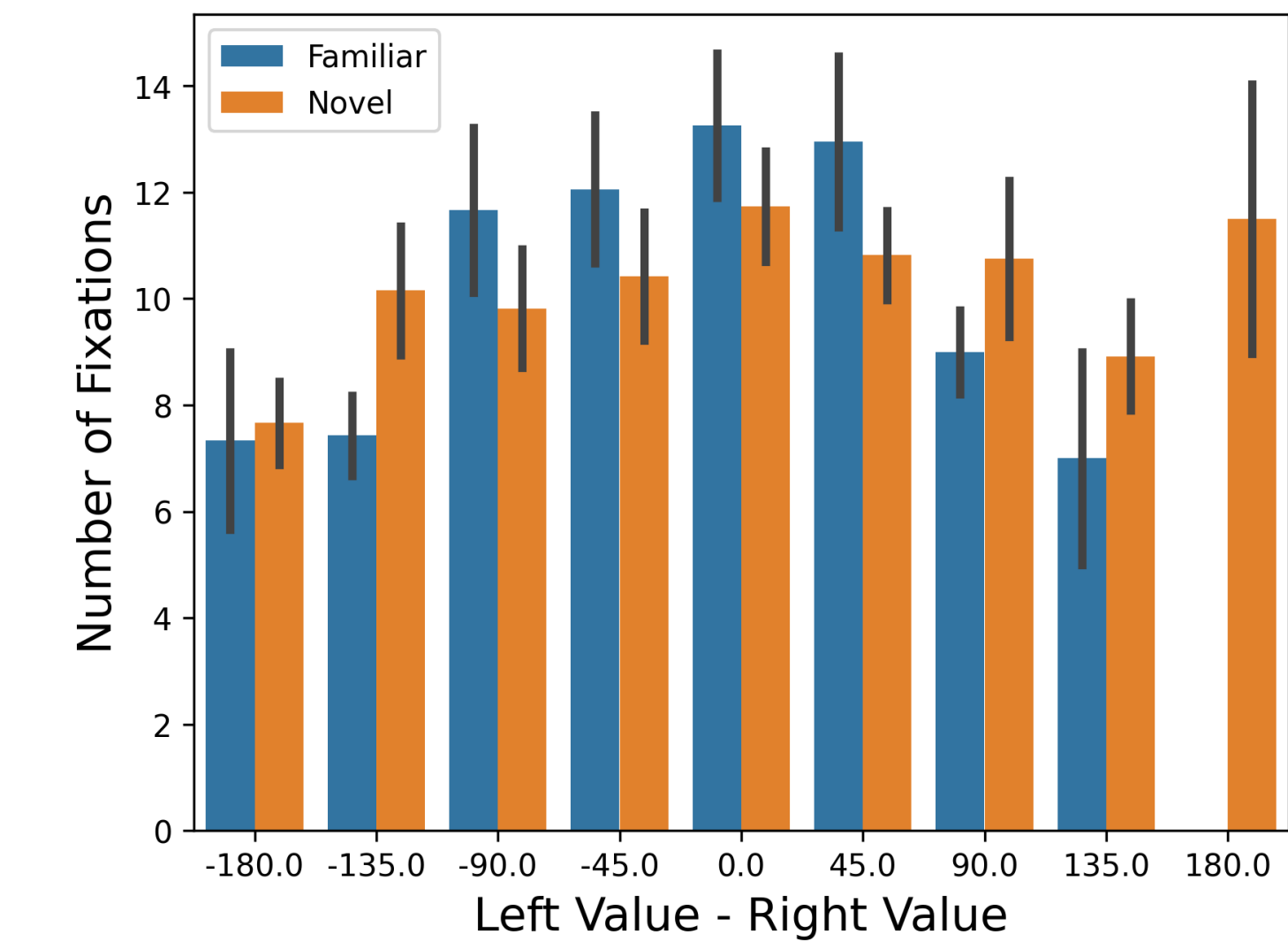
- ▶ Attributes have several levels. 6 levels for central shape, 9 levels for background color and 11 levels for line orientation.
- ▶ Each level is associated with different number of points, which participants learned by trial-and-error in a separate task.
- ▶ Stimuli were split in two conditions:
 - ▶ Familiar: 12 stimuli
 - ▶ Novel: 584 stimuli
- ▶ Stimulus pairs consisted of items from the same condition.
- ▶ Sample: 6 participants
- ▶ Learning task: 8800 trials over 3 weeks
 - ▶ Familiar: 5280 trials
 - ▶ Novel: 3529 trials
- ▶ Example data:

Trial Number	Type	Value Left	Value Right	Left Fixation	Fix Duration	Choice	Response Time
1	Familiar	150	20	1	240	Left	820
1	Familiar	150	20	0	400	Left	820
1	Familiar	150	20	1	180	Left	820
2	Novel	45	75	1	220	Right	2610
2	Novel	45	75	0	630	Right	2610

REFERENCES

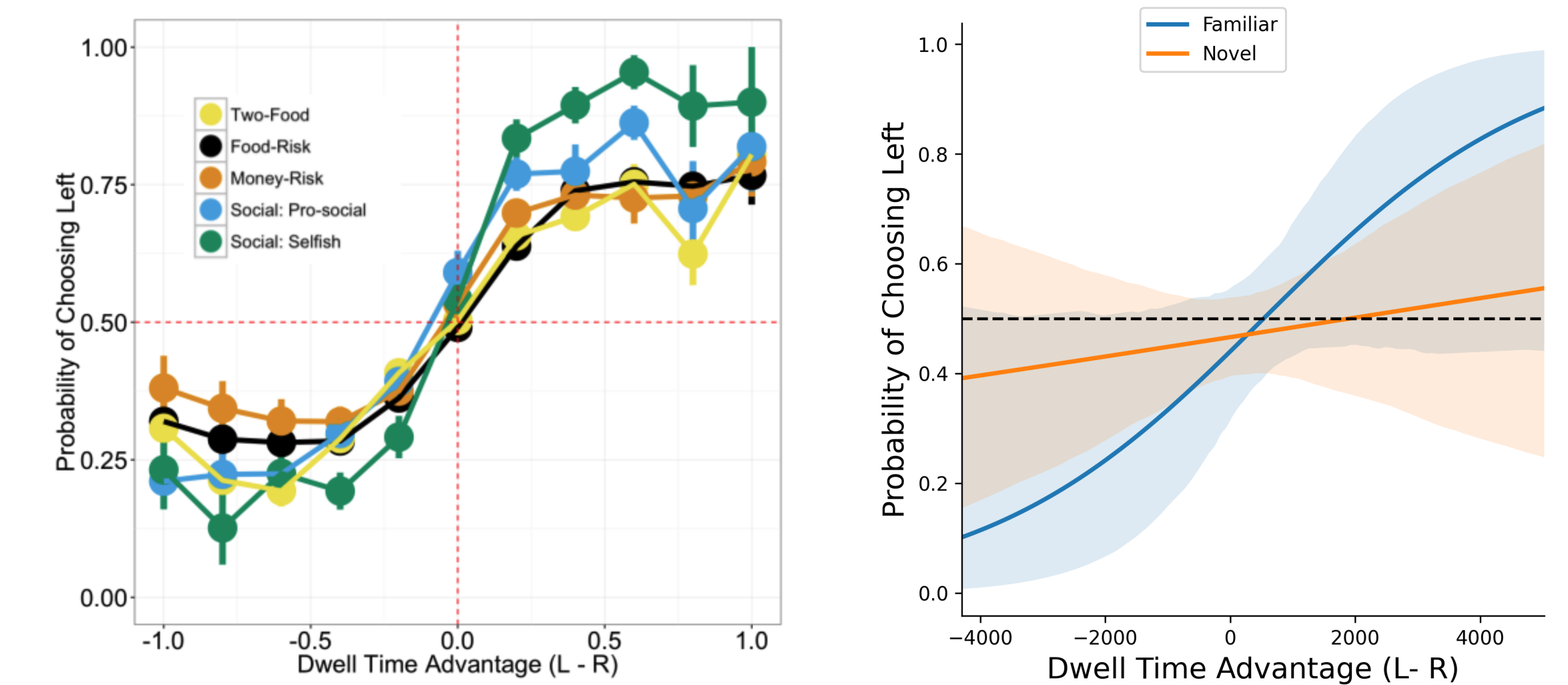
- [1] Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Memory, attention, and choice. *The Quarterly journal of economics*, 135(3):1399–1442, 2020.
- [2] Ian Krajbich, Carrie Armel, and Antonio Rangel. Visual fixations and the computation and comparison of value in simple choice. *Nature neuroscience*, 13(10):1292–1298, 2010.
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EFFECT OF ITEM FAMILIARITY AND CHOICE DIFFICULTY ON ATTENTION



- ▶ Attention and choice duration increases with choice difficulty.
- ▶ Choice difficulty is operationalized as the value difference between the two options.
- ▶ Attention is operationalized as the number of fixations before a decision is made.
- ▶ The effect of choice difficulty on attention is larger for familiar items compared to novel items
 - ▶ The number of fixations for familiar items for the hardest choices is higher compared to novel items and they decrease faster with decreasing difficulty.

EFFECT OF ITEM FAMILIARITY AND ATTENTION ON CHOICE



- ▶ Previous work shows [3] that the more an item is attended to during the deliberation phase, the more likely it is to be chosen.
- ▶ *But* the literature has only examined this effect for familiar items.
- ▶ We find, that attention *only* affects choice for familiar items.
- ▶ Increased attention to a given item does not change the chance of choosing it for novel items.
 - ▶ The amount of attention is operationalized as the difference between dwell times between the options

DISCUSSION

- ▶ The effect of choice difficulty on attention is larger for familiar items compared to novel items.
 - ▶ This could be explained by difference in item value representations depending on familiarity.
 - ▶ The less familiar the options, the noisier the value representation.
- ▶ Attention affects choice only for familiar items not novel items.
 - ▶ This has not been documented in the literature before because most experiments include only familiar items.
- ▶ Our findings suggest potential differences in the cognitive mechanisms underlying value-based decisions that should be clarified in future research.